

The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

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Abstract. Intraday forecasting systems routinely report directional hit-rates between 60% and 85%, and those figures move real capital. We show they are not valid estimates of skill. A k -bar-ahead forecast issued every bar shares $k-1$ of its k outcome bars with its neighbour; when the model's calls also persist, as they do on a live system we instrument, the reported hit-rate acquires a variance far larger than its nominal sample size implies. The reading is not biased upward. Its stated precision is. A forecaster with *exactly zero skill*, simulated against real NSE index bars, posts a $\geq 71\%$ session once in 48 while reporting an interval $1.68\times$ too narrow, rising to $3.40\times$ at the per-minute density these systems publish. We give the closed form behind this, the classical design effect with clusters given by runs of identical calls, and a calibrated interval built on it that covers at 93% where the conventional one covers at 73%. Applied to our own results it leaves nothing: across 48 model-index-horizon tests, 16 clear $p < 0.05$ scored per bar and *zero* clear it per event. Break-even accuracy derived from measured move distributions is 71.5% at thirty minutes against a best achieved 51.1%, and at five minutes no accuracy suffices under a symmetric bracket. Code and raw results are public.

Keywords: Overlapping observations · Backtest overfitting · Directional forecasting · Negative results · Reproducibility · Index derivatives

1 Introduction

A system reporting 71% directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, costs and multiple testing places realistic out-of-sample intraday index accuracy close to a coinflip [1,2], while surveys of the applied literature catalogue many higher figures reported without those controls [3].

This paper began as an attempt to build such a system. Over fourteen months we ran a live multi-horizon forecasting platform for Indian index derivatives, and it reported 71%. That number was wrong. The session producing it cannot be re-audited, the portion of the log holding it having been lost before this analysis; every figure below comes from sessions that survive.

What the overlapping-observations literature does not document is the magnitude of the distortion in a now-ubiquitous practice: the every-minute multi-horizon dashboard, where overlap meets a second ingredient, prediction persistence, and is amplified across a panel.

Contributions. (i) We measure the artefact as it occurs, showing that overlap alone changes nothing and that inflation requires overlap *and* persistent predictions, with persistence calibrated from a production log rather than assumed. (ii) We derive the effective sample size of such a stream, show it is governed by $\mathbb{E}[L^2]/\mathbb{E}[L]$ over run lengths, and calibrate an interval that covers at its nominal rate with no fitted constant. (iii) We apply it to our own results and to a complete control set — shuffled labels, injected positive controls, eight architectures, a block-bootstrap null — and report that nothing survives. (iv) We invert accuracy into a required-versus-achieved framing that shows the gap is not close, and is in places unbridgeable at any accuracy. Code, raw results and the free-data corpus are at <https://github.com/rachitranka25/accuracy-illusion>.

2 Related Work

Hansen and Hodrick [4] characterise inference under overlapping returns and Newey and West [5] give the standard correction, but that work concerns regression coefficients. We apply the same logic to what practitioners publish, the session hit-rate, and reach the classical design effect [6] with clusters given by runs of identical calls. The contribution is the application to forecast streams, not the statistic. Boudoukh et al. [7] are the closest antecedent: they show overlap manufacturing apparent long-horizon predictability that vanishes under correct inference.

On multiplicity, Harvey et al. [8] argue the conventional $t > 2$ threshold is indefensible after decades of factor mining. Two deflation statistics address the concern from different directions: one discounts a reported Sharpe ratio by the number of trials behind it [9], the other estimates how often an in-sample winner fails out of sample [10]. White’s reality check [11] and the false-discovery-rate procedure of Benjamini and Hochberg [12] are complementary, and Sullivan et al. [2] apply the former to a universe of technical trading rules — the direct precedent for our strategy sweep. On machine learning for finance, Fischer and Krauss [13] report performance gross of costs and Gu et al. [14] modest out-of-sample R^2 , while Zhang et al. [15] and Kolm et al. [16] find genuine short-horizon alpha from message-level data rather than bars.

The point is not that any of that work is wrong but that the scoring unit differs: those studies aggregate per day or per month over non-overlapping periods, where the artefact cannot arise, or score per event. Per-bar scoring of a multi-bar horizon, the condition under which it does arise, is characteristic of intraday dashboards rather than of the published work we cite. None of them reports a multiple-testing correction or a shuffled-label null in the form used here. This paper scores per event, corrects by Bonferroni and Benjamini–Hochberg, carries

a shuffled-label null and injected positive controls, models costs explicitly, and releases code and raw output.

3 Data and Method

All data is obtainable free from a retail brokerage account and public exchange archives: 26 265 and 26 267 five-minute bars on BANKNIFTY and NIFTY 50 from February 2025 to July 2026, 9.6k–14k bars on three further indices, 1 382 738 rows of end-of-day option chain with open interest, a ten-year chain history covering 2 181 days, 515 days of participant-wise open interest and 639 days of index futures carrying real volume. One practical note: index spot prints zero volume, so a feature computed as a “VWAP” on spot is an unweighted expanding mean, and real volume exists only on futures.

Every accuracy below passes through one harness. Splits are expanding-window walk-forward with the k bars whose labels overlap the test window purged from training [17]. Accuracies carry Clopper–Pearson [18] intervals, exact binomial p -values and the Pesaran–Timmermann statistic [19]. Every model is additionally trained on randomly permuted labels; if real and shuffled scores coincide, the pipeline is reading noise. A synthetic feature of known strength is injected as a positive control, since a pipeline that cannot recover planted signal cannot be trusted to report a null. Strategy returns use non-overlapping k -bar holds with explicit costs.

Units. All analysis is on 5-minute bars, so the 30-minute horizon is $k = 6$ and a session yields $n = 69$ forecasts. The production system emits a forecast every *minute*; pooled, its run lengths average 15.2 minutes with a longest of 263, and the pooled mean run in 5-minute bars is 3.4.

4 The Mechanism

4.1 Overlap is arithmetic

Measure the autocorrelation of the k -bar forward return two ways. On overlapping windows it reaches 0.841 at lag 1 and decays to approximately zero at *exactly* lag k ; sampled so that no two observations share a bar it is 0.067 at lag 1 and indistinguishable from zero thereafter. The pattern reproduces at $k = 6$ and $k = 12$, with the decay landing on lag 6 and lag 12 respectively. The correlation is a property of the sampling scheme, not of the market, and it is the whole of the mechanism below.

4.2 Persistence is the second ingredient

Overlap alone is insufficient, and getting this wrong was instructive: if predictions are drawn independently each bar the hit indicator is i.i.d. Bernoulli however

autocorrelated the outcome, and our first simulation made that error and correctly found nothing. Real models do not re-decide every bar. Measured on the production log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 25.9 minutes and NIFTY 50 for 10.3, against a mean run of one for a model that re-drew each bar. When both call and outcome persist, one lucky trending stretch is recorded as dozens of hits.

4.3 What a zero-skill forecaster reports

We simulate a fair coin held for a run length drawn from the measured distribution, evaluated against real bars and scored as a dashboard scores it (Table 1).

Table 1. A forecaster with exactly zero skill, 30-minute horizon, scored per bar versus per event. BANKNIFTY; NIFTY 50 agrees within 0.6 points on every row.

	Per bar Per k bars	
Reported sample size per session	69	12
Stated binomial 95% CI	± 11.8 pts	± 28.3 pts
Realised spread across sessions	± 19.9 pts	± 28.0 pts
Understatement factor	$1.68\times$	$0.99\times$
$P(\text{session} \geq 60\%)$	16.2%	18.9%
$P(\text{session} \geq 71\%)$	2.1%	7.1%

The table is easily read backwards. Per-bar scoring does *not* produce extreme readings more often than honest scoring; it produces them *less* often, and reports them with an interval $1.68\times$ too narrow because the quoted n of 69 is a fiction. Overlap manufactures unwarranted *confidence*, not high hit-rates.

4.4 At the real granularity, and across a panel

The production system forecasts every minute, so a session yields about 313 forecasts rather than 69, and simulating the same zero-skill forecaster against the log’s own outcome series raises the understatement factor to $3.40\times$ while extreme sessions become slightly *rarer*. Every 5-minute-bar figure here therefore understates the artefact. A dashboard also watches more than one number: two indices at four horizons is eight hit-rates per session, and simulating the panel jointly gives $P(\exists \text{ series} \geq 71\%) = 17.1\%$ — about one session in six, so a headline 71% arrives within the first weeks of live operation with no edge at all.

5 A Calibrated Estimator

5.1 Effective sample size

Let $h_t = \mathbf{1}\{\hat{y}_t = y_t\}$ and $\hat{p} = n^{-1} \sum_t h_t$. Practitioners quote $\text{Var}(\hat{p}) = p(1-p)/n$, valid only for independent h_t . In general

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[1 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \rho_j \right], \quad (1)$$

with $\rho_j = \text{Corr}(h_t, h_{t+j})$. Because the dependence is block-shaped rather than smoothly decaying, this collapses: if the sequence decomposes into runs of length L_i within which h_t is constant, then

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

This is the classical design effect [6]; what is new is that a forecast stream has exactly this structure and that nobody reports it, and that the inflation is governed by $\mathbb{E}[L^2]$, so the tail matters.

5.2 Validation by coverage

Table 2 reports coverage against a forecaster of known skill whose calls persist per the measured distribution, with true skill measured from the generative process rather than assumed.

Table 2. Coverage of a nominal 95% interval on a session hit-rate. BANKNIFTY, 30-minute horizon, nominal $n = 69$; NIFTY 50 agrees within 2 points. HAC-L gives the Bartlett kernel a run-structure bandwidth; *derived* divides the design effect by the alternation factor of (3).

True skill	Naive	HAC	HAC-L	Design	Derived	n_{eff}
50%	73.4	86.2	88.0	99.7	93.2	11.9
54%	73.9	86.6	88.1	99.6	93.5	11.7
59%	73.0	85.2	86.8	99.6	93.5	10.9

HAC and a block bootstrap (86.2%, omitted from the table) both under-cover by about nine points. The diagnosis is bandwidth: the automatic Bartlett rule gives 3.7 bars at $n = 69$ against a pooled mean run of 3.4 and a tail reaching 53, and a run-structure bandwidth recovers only one to two points. The design effect (2) removes the failure mode entirely and over-corrects in the other direction, reaching 99.7%. That over-statement is derivable rather than incidental: treating runs as independent blocks ignores that a binary sequence’s runs alternate deterministically, a run of hits always being followed by a run of misses. For geometric run lengths of mean μ ,

$$\frac{\mathbb{E}[L^2]}{\mathbb{E}[L]} = 2\mu - 1, \quad \text{whereas exactly } \frac{1 + \rho}{1 - \rho} = \mu - 1, \quad (3)$$

so (2) over-states the dependence by a factor $(2\mu - 1)/(\mu - 1)$, a quantity that tends to 2 for long runs. Evaluated per session it averages 2.44 at a mean hit-run length of $\mu = 3.53$; the residual is the departure of real run lengths from geometry.

Dividing by that factor, computed from each sample’s own mean run length, introduces no free constant and gives 93.1–93.5% coverage across both indices and all three skill levels. We report it as the recommended estimator, and the raw design effect as the conservative bound.

5.3 Reliability, and a real session

Equation (2) depends on a tail-driven second moment estimated from a few dozen runs, and a tail is what a short sample estimates worst: against a pooled reference of 10.6, a single session’s n_{eff} lands between $0.63\times$ and $1.73\times$ that value nine times in ten, so it should be estimated from a run of sessions. That matters immediately. On the production log the best NIFTY 50 session, 69.6% over 335 forecasts, has a naive interval excluding 50% and corrects to $[50.0, 89.1]$, which still excludes it — but at the 5th percentile of the n_{eff} uncertainty it becomes $[45.1, 94.0]$ and no longer does. The margin is smaller than the estimator’s own noise, so the session supports no claim of skill. An earlier draft reported that it did.

Per-event scoring is unbiased and is what we use for every backtest here, but it is insufficient for a live stream: a dashboard cannot wait k bars, subsampling throws away 83% of the observations, and which of the k offsets is chosen moves the same session by 7 to 20 points.

6 Predictability Under Correct Scoring

6.1 Is there anything to predict?

Before asking which model is best, we ask whether any model could succeed. An unconstrained forest was fitted until it memorised the training data, on five indices across two exchanges. Training accuracy reaches 100.0% everywhere and test accuracy stays at chance: 50.5 and 50.0 on BANKNIFTY and NIFTY 50 over 16 856 and 16 853 scored bars, 49.7–50.2 on the other three. Permuting labels moves accuracy by 0.3 to 0.7 points, no real-label result is distinguishable from a coinflip ($p = 0.22$ to 0.91), and both injected positive controls, planted at about 55% and 65%, are recovered at $p < 0.001$ everywhere. The null is a property of the data, not of the estimator.

6.2 Eight model families, three horizons

Eight families were trained on identical splits, features, seeds and controls: logistic regression, gradient boosting, LightGBM [20], XGBoost [21], k -nearest neighbours, an LSTM [22], a Transformer [23] and a compact Temporal Fusion Transformer [24]. Counting families significant at $p < 0.05$, per-bar scoring yields 0 and 2 at 15 minutes on BANKNIFTY and NIFTY 50, 3 and 1 at 30 minutes and 5 and 5 at 60: 16 of 48 tests. At an effective sample size, and again scored one observation per event, the count is *zero* in all six cells.

Scored per bar, XGBoost reaches 51.3% on BANKNIFTY at $p = 0.001$ — in an earlier draft our one surviving positive result, described as robust to Bonferroni correction. It does not survive our own correction: re-scored one observation per 6 bars it is 50.4% at $p = 0.692$, and widening intervals to n_{eff} puts 50% inside every one. The count of *apparent* discoveries rises with horizon, exactly as

it should if the artefact scales with k ; the count of surviving discoveries stays at zero.

The zoo contains its own diagnostic: the NIFTY 50 Transformer sits at 48.4% with $p < 0.001$, which at the nominal n is 4.3 standard errors *below* chance, where the largest $|z|$ across 48 tests should land near 2.9–3.2. That is the variance inflation of Section 4 surfacing in our own results; at n_{eff} it vanishes.

6.3 Eighty-six strategies

Separately, 86 parameterised rules — momentum, mean reversion, moving-average crosses, RSI, Bollinger, breakout, gap, volatility-regime, time-of-day and streak variants — were evaluated on non-overlapping 30-minute holds at 4 bp cost with a one-sided test for profit. Two have a positive Sharpe ratio; none is significantly profitable, before or after Benjamini–Hochberg.

The deflated Sharpe ratio returns an implausible null benchmark of 7.77 here, because trial Sharpe dispersion is dominated by deterministic turnover differences rather than by noise, violating the assumption it rests on. We replaced it with a circular block bootstrap [25] that resamples forward returns while holding each strategy’s positions fixed: the best observed Sharpe is 0.21 against a null best-of-86 median of 0.44, with $P(\text{null} \geq \text{observed}) = 0.658$. The winner of the sweep is worse than what selection alone produces from no edge at all. The probability of backtest overfitting via combinatorially symmetric cross-validation [10] is degenerate here and reported as such.

7 Economic Significance

A hit-rate without a payoff ratio is not a result. For a symmetric bracket at the typical move m with round-trip cost c , expectancy is $m(2p - 1) - c$, so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (4)$$

The ratio c/m is the entire problem. At 4 bp round-trip cost, with the median move taken from seventeen months of bars, p^* is 79.8%, 71.5% and 65.4% at 15, 30 and 60 minutes on BANKNIFTY and 84.9%, 75.1% and 67.7% on NIFTY 50, against best per-event accuracies of 51.3%, 51.1% and 52.2%, and 50.5%, 51.2% and 50.1%. A hold-to-horizon rule, which collects the mean move rather than the median, lowers the requirement to 71.3–60.8% and 74.9–62.7% and does not close the gap either.

At the shortest horizon the median move is smaller than the round trip, so under a symmetric bracket p^* exceeds 100% and no accuracy suffices. That is specific to the bracket: moves are right-skewed, so a hold-to-horizon rule requires 86% and 93% instead — far beyond anything reported here or in the literature we cite.

Second, instrument choice can make the requirement unreachable. On the BANKNIFTY 30-minute horizon, where the median move is 9.3 bp, break-even

is 55.4% at 1 bp of cost, 71.5% at 4 bp and 93.0% at 8 bp; at the 20 bp a retail at-the-money option round trip costs over that horizon, $c > m$ and $p^* > 1$, so a perfect forecaster still loses on the median trade. The instrument most retail participants use, over the horizon they hold it, has no directional accuracy that makes it profitable.

A genuine 85% would clear break-even by 13.5 points on BANKNIFTY futures and still lose on options: such a claim is either an artefact of the kind measured in Section 4 or true and far less valuable than it sounds.

8 What Is Predictable Instead

These results went through the same harness, and what it leaves is narrow enough to state precisely: *none of this project's findings survives as a finding of ours.*

Volatility, confirmed but not ours. A binary magnitude target reaches 70.8% per bar and 71.3% per event on BANKNIFTY (72.0 and 72.1 on NIFTY 50), with shuffled controls at 52–54%. That it survives the correction which destroyed the directional result matters, since an instrument flattening everything would be useless. But a persistence rule already scores 67.4%, the edge is +3.3 to +3.7 points per event, and at n_{eff} the two intervals overlap: this confirms volatility clustering [26] rather than discovering anything.

Premium selling and positioning, withdrawn. Short at-the-money straddles held to expiry, on real end-of-day premiums across four entry points, are not significantly profitable on either index before or after Benjamini–Hochberg, and no bootstrap interval on the mean excludes zero on the profitable side; recentring and resampling gives a best-of-four null median of 17.8 against an observed 12.1 on NIFTY ($P = 0.68$). The exchange also moved BANKNIFTY options from weekly to monthly expiry mid-sample, so the indices were never comparable: split on expiry spacing the weekly sub-sample returns –67.5 points per trade and the monthly +57.5. The premium is real and documented [27]; our measurement of it as an edge is withdrawn. Option-chain positioning fares no better: over 2181 days, six of eight price-residual features clear $p < 0.05$ uncorrected and *one* survives Benjamini–Hochberg — the daily change in the put–call open-interest ratio ($IC = 0.059$, $t \approx 2.8$), also the only feature strengthening in the second half. Nothing in the price-only set clears $p < 0.05$ at all.

9 Limitations

The alternation correction assumes geometric run lengths; real runs depart from that, which is why coverage lands near 93% rather than 95%. Two indices carry the primary intraday analysis over seventeen months in one market; the ceiling test replicates across five, but the model zoo and strategy sweep do not. Break-even is computed for two payoff models, neither of which models the price path. Two figures that were re-run did not reproduce and are withdrawn in place

rather than quietly deleted, and our Temporal Fusion Transformer is a compact variant, not the published architecture [24]. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [16,15], is untested here for want of message-level history.

10 Conclusion

The headline metric of intraday directional forecasting is not merely noisy; its stated precision is systematically wrong, by enough to make a coinflip look like a discovery. Applied to our own work the correction leaves nothing standing. Sixteen of 48 model-index-horizon tests clear $p < 0.05$ per bar and none clears it per event; 86 strategies produce a winner worse than a bootstrap null; premium selling, published as this project’s one measurable edge, fails the same correction; and the one untouched result confirms volatility clustering rather than finding anything new.

The honest summary is that this project found nothing tradeable, and its value lies in how thoroughly that was established. What we offer is a method: an effective sample size for overlapping persistent forecasts, a calibrated interval built on it, and a framing that turns an accuracy figure into a question about cost. Scoring unit, effective sample size, multiplicity correction, a shuffled-label control and an explicit cost model are not optional refinements: without them an accuracy figure means nothing, and we know that because we published several that did not.

Reproducibility, data availability and scope. Every number here is re-generated by a script in the public repository, with the raw output of each committed alongside it, at <https://github.com/rachitranka25/accuracy-illusion>. That repository holds the project’s complete research record, including the positive results withdrawn above; all experiments use a fixed seed, and collectors for every data source are included so the corpus can be rebuilt. This is a measurement study, not investment advice, and its central finding is one we apply first to our own published claims.

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